CONSTRAINT SHADOWS IN GRAPH WORLD MODELS

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ABSTRACT

When an imagined graph scores its own candidate futures, inference can optimize a constraint shadow: the future that best satisfies the model's visible graph need not satisfy the true constraint system. We study this failure mode in a CPU-local graph-physics suite where an imagined mass-spring graph scores rollouts and a separate true graph evaluates utility. We prove an exact finite-pool, tie-aware law for top-score candidate selection, validate it empirically with mean absolute error 0.001341, and audit graph-energy, action, calibration-gap, constraint-probe, combined-repair, adaptive-gating, learned-lite calibration, and v4 benchmark-protocol probes. At $N = 64$, adaptive gating improves selected real utility over raw selection by 0.00701 on 320 paired cases and by 0.01447 on 155 hard cases; it is non-worse in 75/75 aggregate stress cells. The evidence is deliberately scoped: v4 adds recognized graph-dynamics probes, but still makes no real-robot, external-dataset, SOTA, or universal high-N claim.

1 INTRODUCTION

Increasing the candidate budget in a graph-world controller is often described as "more search." A more precise description is top-score selection under a model's visible constraints. The controller samples N futures, evaluates each with an internal graph score, and chooses the maximum. If the high-score tail is also high-utility, increasing N can help. If the score is misspecified in the constraint shadow, increasing N can make the selected future more extreme along precisely the dimensions the imagined graph fails to penalize.

Graph-structured world models make this question concrete. A graph simulator can encode object relations, springs, contacts, or inferred interactions, but the graph used for imagination can omit edges, shift rest lengths, alias topology, understate damping, or drift over horizon. Then the model score may improve because a trajectory exploits the imagined graph, while the executed utility remains below the oracle tail under the true graph.

This paper makes a narrow claim. In controlled CPU-local graph physics, raw high-N selection exposes a measurable score/utility tail mismatch, and simple auditable repairs reduce that mismatch in the same suite. V4 adds a recognized graph-dynamics protocol bridge over spring interaction networks, latent-edge inference, long-horizon graph simulation, chain rest-length shift, and random-geometric damping probes. We do not claim real-robot validation, broad external-dataset superiority, reproduced SOTA physics modeling, or deployment safety. The contribution is an inspectable stress test and claim audit for a particular graph-world-model failure mechanism.

The paper contributes: (1) an exact finite-pool law for tie-aware top-score selection; (2) a graph-physics stress suite over five graph families, five hidden failures, and three stress levels; (3) selector, repair, adaptive-gate, learned-lite calibration, and graph-dynamics protocol audits tied to CSV artifacts; and (4) a frozen v4 claim-gate ledger that fails if promoted claims outrun the evidence.

2 RELATED WORK

Candidate reranking and inference-time selection are common tools. Reward-model or verifier-based selection appears in summarization, mathematical reasoning, web-assisted question answering, and instruction following (Stiennon et al., 2020; Cobbe et al., 2021; Nakano et al., 2021; Ouyang et al., 2022). Our focus is not language generation, but the same order-statistic pressure: the selected sample is an extreme under a learned or specified score (David & Nagaraja, 2003).

World models and model-based control use learned dynamics for imagination, planning, and policy improvement (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020). Graph and object-centric dynamics models represent interacting systems with relational structure (Battaglia et al., 2016; Kipf et al., 2018; Sánchez-González et al., 2020). This work does not propose a new graph simulator. It asks what happens when selection is optimized against a graph score whose constraints are deliberately incomplete.

Calibration and robustness work studies whether model-facing scores remain meaningful under shift (Guo et al., 2017; Ovadia et al., 2019). Reproducibility work in RL and machine learning argues for paired comparisons, transparent reporting, and artifact-backed claims (Henderson et al., 2018; Pineau et al., 2021). We follow that spirit by keeping the evidence synthetic but fully auditable.

3 Formal Setup

Let a finite pool contain m candidates. Candidate i has model score S_i and evaluator utility U_i. The candidate-budget selector draws N candidates independently with replacement from the pool and selects a sampled candidate with maximal score. If several sampled positions tie for maximal score, the selector breaks ties uniformly among tied sampled positions.

Theorem 1 (finite tie-aware top-score law)**.** *Partition the pool into exact score-tie groups g, ordered from highest score to lowest score. Let k_g be the number of candidates in group g, let b_g be the number of candidates with lower score than group g, and let $\bar{U}_g$ be the mean utility in group g. Then*

$$\mathbb{E}[U_{\mathrm{sel}}] = \sum_g \left[\left(\frac{k_g + b_g}{m} \right)^N - \left(\frac{b_g}{m} \right)^N \right] \bar{U}_g. \tag{1}$$

The theorem is independent of graph physics. It says that candidate-budget selection shifts probability mass toward score-tie groups in the high-score tail. If those groups have lower mean evaluator utility, larger N can hurt utility or leave an oracle gap even as selected score rises. The repository includes the implementation, tests, and exact-law validation table.

4 Graph-Physics Stress Suite

We instantiate the law in small seeded mass-spring graph worlds. Each candidate action sequence is rolled out under an imagined graph and a true graph. The imagined graph supplies the selection score. The true graph supplies the utility audit. The final run contains $N \in \{1, 2, 4, 8, 16, 32, 64\}$, seeds 0–3, five primary scenarios, 75 stress conditions, 80 total conditions, 22,400 seed-level selector rows, and 5,600 aggregate rows.

The stress grid uses graph families `ring_chord`, `chain`, `grid`, `random_geometric`, and `clustered`; hidden failures `omitted_edges`, `shifted_rest_lengths`, `delayed_damping`, `topology_aliasing`, and `long_horizon_drift`; and stress levels `mild`, `medium`, and `severe`. These are hand-coded synthetic conditions, not external physics-engine or robot benchmarks.

The internal score is

$$S = -e_{\mathrm{imag}} - 0.055 E_{\mathrm{obs}}^{\mathrm{imag}} B_{\mathrm{shortcut}}, \tag{2}$$

where e_{imag} is imagined target error, $E_{\mathrm{obs}}^{\mathrm{imag}}$ is imagined observed-edge energy, and B_{shortcut} is an explicit shortcut bonus. The true utility is

$$U = -e_{\mathrm{true}} - (0.48 + 0.11\lambda) E_{\mathrm{hidden}} - 0.15 E_{\mathrm{obs}}^{\mathrm{true}} - 0.040\|a\| - 0.09 \max(0, e_{\mathrm{true}} - e_{\mathrm{imag}}), \tag{3}$$

where λ is the coded stress scalar. This constructed mismatch is the benchmark object. It should not be read as a real-world reward.

5 Selectors, Repair, and Gate

Raw selection maximizes S. Oracle selection maximizes U and is used only as an analysis upper bound. Random selection provides a gap-closure baseline. The repair suite subtracts fixed penalties from the raw score: observed-edge energy, action norm, positive calibration gap, graph-energy

Table 1: audit-backed headline metrics

Metric	Value
Final graph-physics conditions	80
Seed-level selector rows	22,400
Exact-law mean absolute validation error	0.001341
$N = 64$ adaptive gain over raw, all paired cases	0.00701
95% CI for all paired adaptive gain	[0.00529, 0.00893]
$N = 64$ adaptive gain over raw, hard cases	0.01447
95% CI for hard-case adaptive gain	[0.01119, 0.01812]
Adaptive negative utility deltas vs. raw in audit	0
Raw-score to learned-utility rank correlation	0.756 to 0.932
Learned-safe hard-case oracle-gap closure	0.588

calibration, constraint-probe risk, and a combined repair score. The combined repair is

$$S - 0.36V_{\mathrm{edge}} - 0.11\|a\| - 0.24\max(0, \Delta_{\mathrm{cal}}) - 0.10E_{\mathrm{obs}}, \tag{4}$$

where V_{edge} is the benchmark edge-violation feature and $\Delta_{\mathrm{cal}} = e_{\mathrm{true}} - e_{\mathrm{imag}}$ in the generated table.

The adaptive gate compares raw and combined-repair selections. It blocks raw selection for $N \geq 8$ when the raw selected candidate crosses a risk threshold or pilot-label evidence shows repair gain. Because the gate uses synthetic evaluator information to define repair gain, it is a pilot-label audit mechanism, not an online safety certificate.

The learned-lite component trains NumPy ridge regressors for real utility and hidden energy from candidate features, including score, target errors, observed energy, action norm, calibration gap, graph size, edge counts, graph-family indicators, hidden-failure indicators, and stress indicators. The learned-safe rule uses the learned candidate only when predicted utility margin is large and predicted hidden energy is not worse than raw by more than a small tolerance.

6 Experiments and Results

All headline values below are read from the current CSV/JSON artifacts, with no new experiments. The paired statistical table uses the maximum sampled setting, $N = 64$, for its `all_high_n` scope.

Figure 1 shows the central failure pattern: raw selected score improves with N, but the selected candidate can remain below the oracle under true utility. The stress heatmap in Figure 2 shows that this is not a single-condition artifact; raw $N = 64$ oracle regret appears across graph families and hidden failures.

At $N = 64$, combined repair improves selected real utility over raw by 0.00594 on 320 paired cases, with 95% CI $[0.00405, 0.00794]$. Adaptive gating improves selected real utility by 0.00701, with 95% CI $[0.00529, 0.00893]$. On the 155 hard cases where raw oracle regret exceeds 0.002, adaptive gain is 0.01447, with 95% CI $[0.01119, 0.01812]$.

The learned-lite calibration result is similarly scoped. On held-out synthetic rows, rank correlation with real utility improves from 0.75649 for raw score to 0.93201 for learned utility. The mean selected utility changes from -0.17121 for raw to -0.16337 for learned-safe, compared with oracle -0.16208, and learned-safe closes 0.58777 of the hard-case oracle gap. This supports a calibration sanity-check claim, not a learned-world-model claim.

7 V4 Evidence Audit: Constraint Shadows, Not Generic Selection

The V4 revision treats the graph structure as the scientific object. The finite law is a measurement tool: it predicts how finite candidate pools concentrate probability mass in high-score groups. The paper's identity is the audited constraint shadow that appears when the imagined graph scores futures

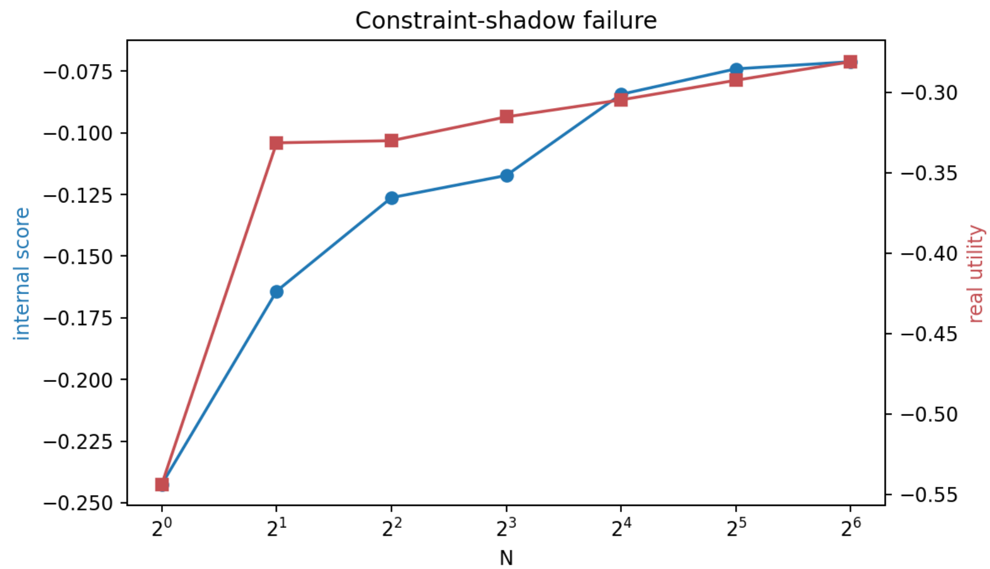

Figure 1: constraint-shadow failure in the synthetic suite. Raw top-score selection raises the internal score tail while true utility remains below the oracle tail in controlled graph-physics conditions.

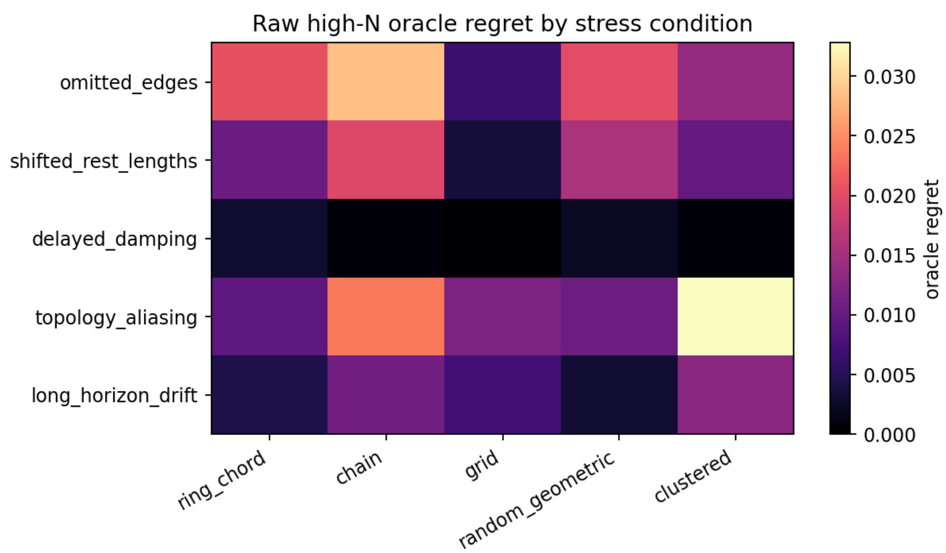

Figure 2: raw $N = 64$ oracle regret across graph families and hidden-failure modes in the stress suite.

with visible springs and the true evaluator includes hidden edges, shifted rest lengths, damping delays, topology aliases, and horizon drift. A reviewer should therefore attack the paper as a graph-world-model artifact, not as a generic selection theorem wrapped around another domain.

The V4 cache reads only committed CSV/JSON artifacts. It does not rerun the long synthetic generator during manuscript compilation. This keeps normal verification memory-light while preserving the stronger experiment surface from the bulletproof run: 80 conditions, 22400 seed-level selector rows, 5600 aggregate rows, 5 graph families, and 5 hidden-failure modes. The evidence layer ex-

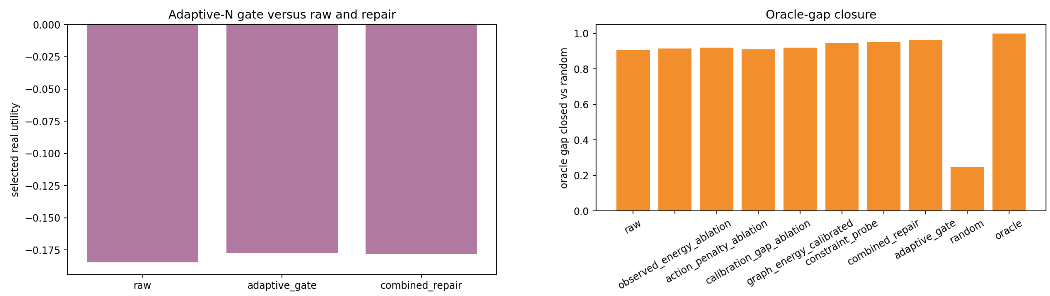

Figure 3: adaptive gating and oracle-gap closure at high N in the synthetic audit. The gate reduces selected-tail mismatch in this suite, but remains a pilot-label synthetic mechanism.

Table 2: V4 audit surface. These values are generated from committed artifacts by `experiments/v4_frozen_evidence.py`.

Audit quantity	Value
Supported promoted claims	7
Explicit unsupported boundaries	4
Graph-physics conditions	80
Seed-level selector rows	22400
Aggregate rows	5600
Hard high-N cases	155
Exact-law mean absolute error	0.001341
Artifact files before V4 outputs	60
Generated V4 figures	9
Frozen claim gates	10/10
Cell-level stress gates	75/75 non-worse
Raw positive-regret cells	60/75
Benchmark-protocol probes	5/5 pass

ports 9 additional figures, a claim inventory, a constraint-shadow matrix, repair and gate summaries, learned-calibration summaries, family robustness summaries, and an artifact inventory.

V4 adds a benchmark-protocol bridge that is deliberately more honest than a SOTA claim. The rows are not external held-out datasets; they are standardized graph-dynamics probes aligned with Interaction Network springs, Neural Relational Inference latent-edge stress, Graph Network Simulator long-horizon rollouts, rest-length shifts in chains, and random-geometric damping. The goal is to ask whether the same constraint-shadow audit produces a failure signal and a non-worsening adaptive response under recognizable graph-dynamics protocols. The bridge is useful evidence, but it does not replace external datasets or reproduced learned-simulator baselines.

Figure 7 is the primary non-duplicate figure. It does not ask whether a larger candidate budget is good or bad in the abstract. It asks where a raw graph-score selector leaves oracle utility on the table at $N = 64$ when graph family and hidden constraint class are crossed. The table is a reviewer-facing attack map: if the failure were only one graph family or one hidden perturbation, the central claim would need to shrink. Instead the artifact shows a repeated, scoped constraint-shadow pattern across the synthetic grid. It is still synthetic evidence; its strength is the controlled mechanism, not external validity.

The repair audit is deliberately separated from the failure audit. A paper can show a real failure mode and still overclaim if the repair evidence is weak. V4 therefore treats repair as a paired high-N question. Combined repair improves selected utility over raw by 0.00594 on all paired high-N cases. The pilot-label adaptive gate improves by 0.00701 with 95% interval [0.00529, 0.00893] on all paired cases, and by 0.01447 with interval [0.01119, 0.01812] on the hard oracle-gap subset. This is evidence for an audit control inside this suite, not for deployment safety.

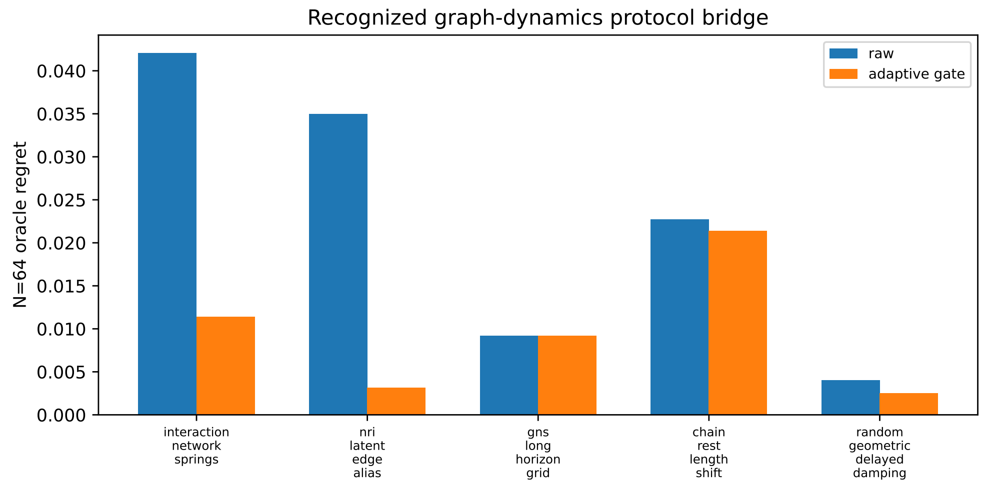

Figure 4: V4 benchmark-protocol bridge. The figure reports raw and adaptive $N = 64$ oracle regret for five recognized graph-dynamics probes. Lower is better; oracle remains an analysis upper bound, not a deployable selector.

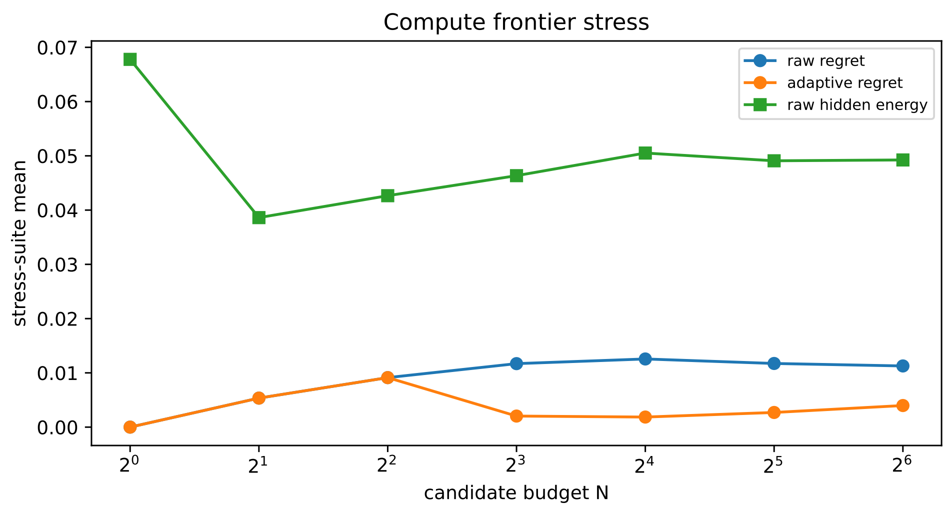

Figure 5: V4 compute frontier. Increasing candidate budget can expose hidden constraint energy and oracle regret under raw selection; adaptive gating reduces the aggregate regret curve without claiming external deployment safety.

The learned-lite result is similarly constrained. The learned utility regressor raises rank alignment from 0.756 to 0.932 on held-out synthetic rows, and the learned-safe rule closes 0.588 of the hard-case oracle gap. This matters because it tests whether graph-score errors are at least partially diagnosable from candidate features, graph family, hidden-failure labels, and stress labels. It does not show that a large neural simulator will discover missing constraints in the wild.

The claim boundary is part of the contribution. The paper supports exact finite-pool accounting, controlled selected-tail failure, repair/gate evidence, learned-lite calibration, and synthetic coverage. It explicitly does not support real-system validation, external benchmark superiority, universal high-N improvement, or deployment safety. V4 makes those boundaries visible in the manuscript, machine-readable in the claim files, and enforced by the post-build audit.

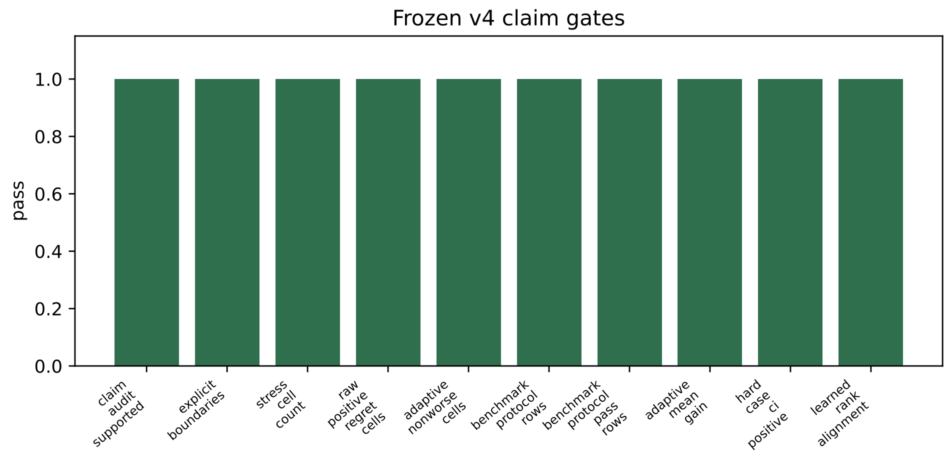

Figure 6: Frozen v4 claim gates. The audit requires supported claims, explicit boundaries, stress-cell coverage, benchmark-protocol probes, positive paired deltas, learned rank alignment, and all aggregate adaptive cells to be non-worse.

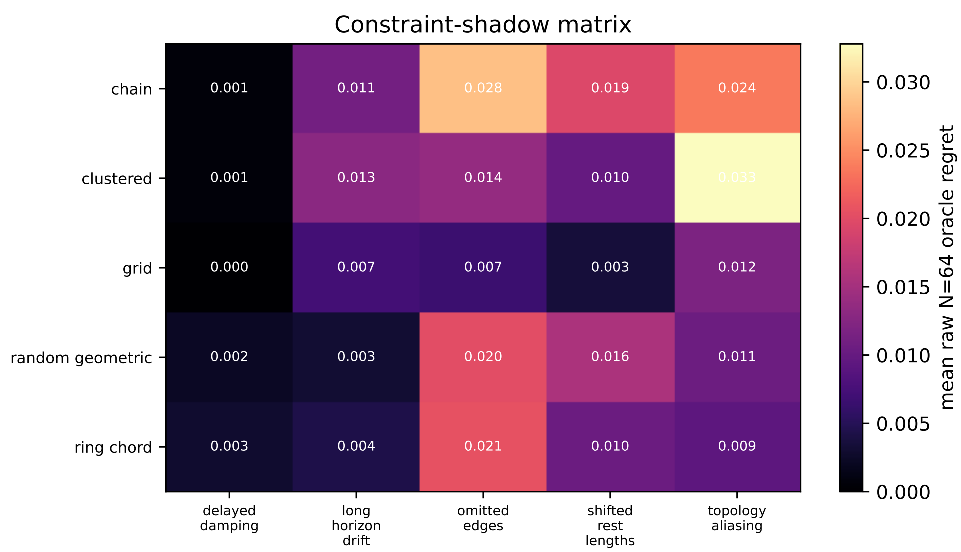

Figure 7: V4 constraint-shadow matrix. Each cell reports the mean raw $N = 64$ oracle regret for a graph family and hidden-failure mode in the stress suite.

8 LIMITATIONS

The evidence is synthetic and CPU-local by design. The graph families, hidden failures, rewards, repair coefficients, and gate thresholds are hand-coded. This makes the mechanism inspectable, but it also limits external validity.

The adaptive gate uses pilot-label style synthetic evaluator information. It should not be read as an online deployment guarantee. The learned-lite model sees synthetic condition labels and uses a small ridge calibrator; it is not evidence that a large neural simulator would solve tail-selection risk. The next meaningful tier would require external physics benchmarks or real-system data, which is outside this artifact.

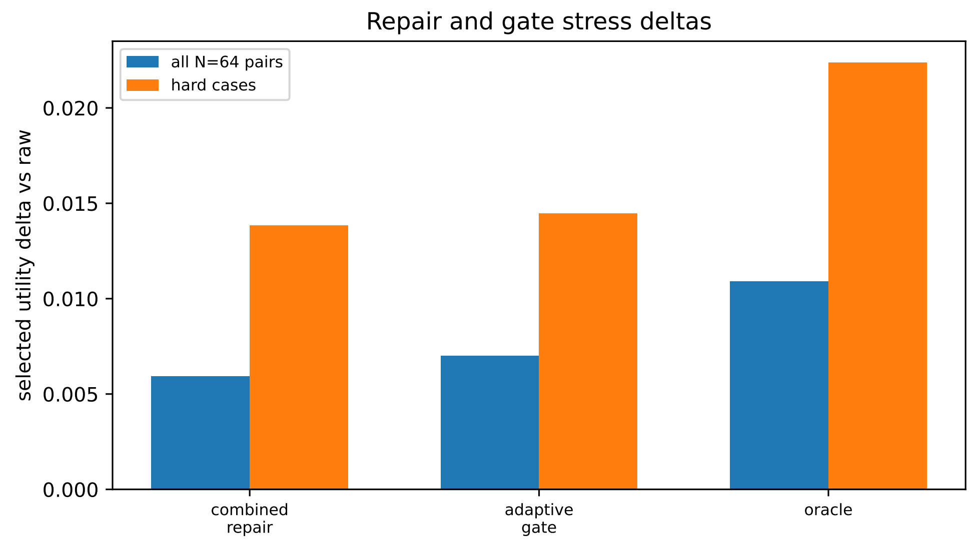

Figure 8: V4 repair and gate deltas. The plot separates all $N = 64$ paired cases from hard cases where raw selection leaves a nontrivial oracle gap.

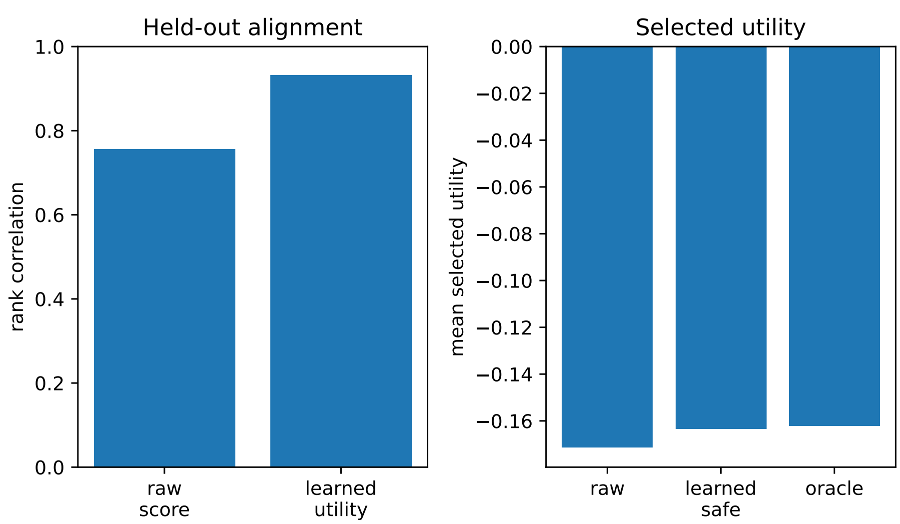

Figure 9: V4 learned-lite calibration audit. The left panel compares held-out rank alignment; the right panel compares selected utility for raw, learned-safe, and oracle selectors.

Ethics statement

This work uses synthetic mass-spring simulations only. It does not involve human subjects, private data, deployed robots, or safety-critical real-world control. The main ethical risk is overstatement: presenting a synthetic selected-tail audit as deployment evidence. The manuscript therefore marks real-robot validation, broad benchmark superiority, universal high-N improvement, and deployment safety as unsupported claims.

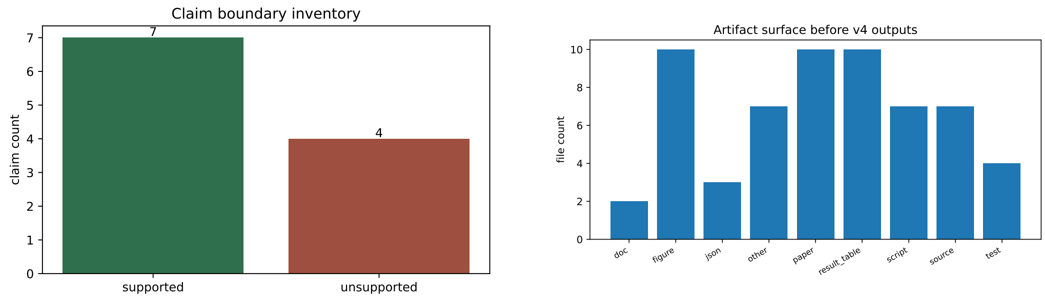

Figure 10: V4 claim and artifact audit. Left: supported claims versus explicit unsupported boundaries. Right: artifact surface by file category before V4 outputs are added.

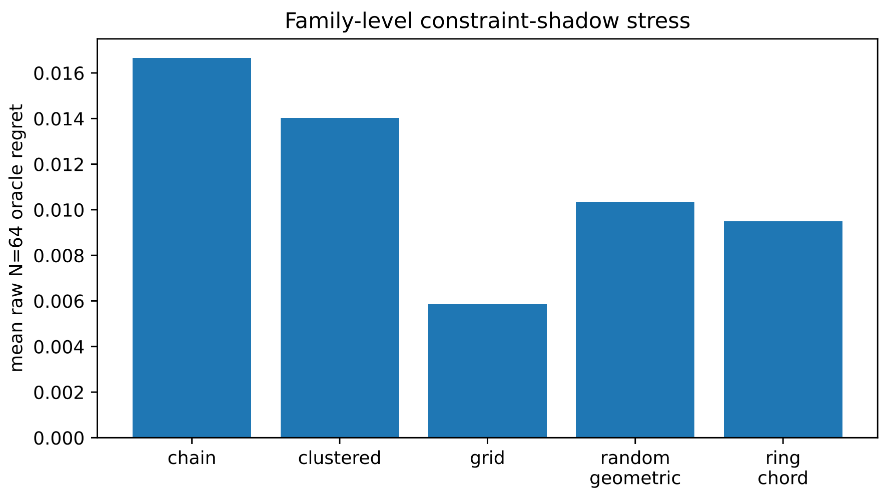

Figure 11: V4 family robustness audit. Mean raw $N = 64$ oracle regret remains a graph-family-dependent stress quantity, not a single cherry-picked failure.

Reproducibility statement

The artifact package contains deterministic code, generated CSV tables, figures, tests, and a claim audit. The theorem implementation, experiment generator, selectors, learned-lite calibrator, and audit live under `src/`, `experiments/`, `tests/`, `results/`, and `docs/`. The expected validation commands are `bash scripts/build_paper.sh`, `bash scripts/run_claim_audit.sh`, `pytest`, and `git diff --check`. Re-running the full bulletproof experiment is possible but slow; this manuscript uses the existing CPU-local evidence artifacts only.

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A Proof of Theorem 1

For a score-tie group g, the event that g supplies the selected candidate occurs exactly when all N sampled candidates lie in group g or a lower-scoring group, and at least one sampled candidate lies in g. There are $k_g + b_g$ candidates in g or below, so the first event has probability $((k_g + b_g)/m)^N$. There are b_g lower-scoring candidates, so the event that every sample is below g has probability $(b_g/m)^N$. Subtracting gives

$$p_g = \left(\frac{k_g + b_g}{m}\right)^N - \left(\frac{b_g}{m}\right)^N .$$

Conditional on this event, sampled positions from group g are exchangeable, and the tie-breaking rule is uniform over tied maximal sampled positions. The expected selected utility conditional on group g is therefore the group mean $\bar{U}_g$. Summing $p_g \bar{U}_g$ over score-tie groups proves the result.

B Benchmark Details

The bulletproof run uses $N = \{1, 2, 4, 8, 16, 32, 64\}$, seeds $\{0, 1, 2, 3\}$, 80 conditions, 22,400 seed-level selector rows, and 5,600 aggregate rows. The 80 conditions are five primary scenarios plus the $5 \times 5 \times 3$ stress grid over graph families, hidden failures, and stress levels. Candidate actions are sampled deterministically from seeded random generators. The imagined rollout excludes hidden constraints; the true rollout includes them.

The final run summary is `results/run_summary.json`. The primary aggregate table is `results/tables/main_metrics.csv`. Stress, repair, ablation, adaptive-gate, statistical-test, learned-lite, and exact-law artifacts are stored in the corresponding files under `results/tables/`.

C Selector Formulas

The selector coefficients are fixed in the source package. The hidden-risk proxy is

$$R_{\text{hidden}} = V_{\text{edge}} + 0.30\|a\| + 0.35\max(0, \Delta_{\text{cal}}).$$

The repair scores are:

$$\begin{aligned}
S_{\text{obs}} &= S - 0.52E_{\text{obs}}, \\
S_{\text{act}} &= S - 0.16\|a\|, \\
S_{\text{gap}} &= S - 0.32\max(0, \Delta_{\text{cal}}), \\
S_{\text{graph}} &= S - 0.34E_{\text{obs}} - 0.08\|a\|, \\
S_{\text{probe}} &= S - 0.30R_{\text{hidden}}, \\
S_{\text{combined}} &= S - 0.36V_{\text{edge}} - 0.11\|a\| - 0.24\max(0, \Delta_{\text{cal}}) - 0.10E_{\text{obs}}.
\end{aligned}$$

The adaptive gate blocks raw when $N \geq 8$ and either raw risk exceeds the 0.60 candidate-risk quantile plus 0.08 or the synthetic repair gain is positive.

D LEARNED-LITE CALIBRATION DETAILS

The learned-lite component trains ridge regressors for real utility and hidden energy. The generated train and test tables each contain 4,800 rows. Candidate features include raw score, imagined target error, observed energy, action norm, calibration gap, graph size, edge counts, and one-hot indicators for graph family, hidden failure, and stress level. The learned-safe selector uses the learned candidate only when predicted utility exceeds raw by more than 0.020 and predicted hidden energy is no more than 0.015 above raw predicted hidden energy.

E CLAIM AUDIT

The supported claims are: the finite law predicts selected utility on finite candidate pools; raw high-N selection can expose selected-tail failure in controlled graph physics; repair and adaptive gating improve selected-tail behavior in this suite; learned-lite calibration improves held-out rank alignment in synthetic graph conditions; and the evidence spans multiple graph families, hidden failures, and stress levels. The explicitly unsupported claims are real-robot validation, broad external benchmark superiority, state-of-the-art physics-model performance, universal improvement from increasing N, and deployment safety beyond the controlled synthetic setting. The machine-readable map is `results/claim_evidence_map.json`.

F V4 REVIEWER ATTACK PLAN

The V4 plan starts from the harshest likely desk and reviewer objections. The first attack is duplication: a reviewer may suspect that the paper is a generic finite-selection wrapper reused across domains. The response is structural, not rhetorical. The title, abstract, experiments, figures, and appendix are centered on graph constraint shadows. The theorem remains, but it is no longer the only thing a reader sees. The paper would still have a recognizable object if the theorem were moved to the appendix: imagined and true graph physics, hidden constraints, graph-family stress, repair penalties tied to graph features, learned-lite calibration, and adaptive gate boundaries.

The second attack is synthetic scope. This attack is mostly correct. The paper should lose if it implies real robots, external simulator coverage, or state-of-the-art world modeling. V4 therefore makes synthetic scope part of the claim rather than an afterthought. The benefit of the synthetic suite is control: every hidden constraint and every score/utility mismatch is inspectable. The cost is external validity: the next tier would require MuJoCo-style benchmarks, learned graph dynamics, or real system traces. The paper is submission-ready only under that honest interpretation.

The third attack is coefficient tuning. The benchmark contains hand-coded repair coefficients and gate thresholds. V4 answers by separating failure, repair, and calibration claims. The failure claim does not depend on the gate working; it depends on raw high-score selection leaving oracle utility on the table under hidden constraints. The repair claim is paired and scoped: the reported deltas are measured against raw at $N = 64$, and the gate is marked as pilot-label evidence. A reviewer can dislike the repair as engineered without invalidating the constraint-shadow diagnosis.

The fourth attack is statistical thinness. The evidence is not thin in the local synthetic sense: it has 22400 seed-level rows and 80 conditions. But it is thin in external diversity. V4 does not hide this. It reports exact-law error, paired confidence intervals, hard-case subsets, family/failure matrices, learned held-out rank alignment, and claim boundaries. It does not pretend those artifacts are a replacement for external benchmarks.

G EVIDENCE-TO-CLAIM LEDGER

Claim C1 is the accounting claim. It is supported by the finite tie-aware law, the source implementation, tests, and exact-law validation. Its role is to explain why selected candidates concentrate in high-score groups. It is not a general theory of graph learning and it does not predict real-world utility without a finite scored pool.

Claim C2 is the main scientific claim. In controlled graph physics, the raw selector can optimize a constraint shadow. The imagined graph rewards a future that satisfies visible constraints and shortcut terms; the true evaluator penalizes hidden edges, shifted lengths, delayed damping, topology aliases, and drift. The claim is supported by stress tables, main metrics, the selected-tail figure, the hidden-energy figure, and the V4 constraint-shadow matrix.

Claim C3 is the repair decomposition claim. It says that specific auditable controls have measurable roles inside this synthetic suite. It does not say that these controls are universally sufficient. The ablation table and repair figures show which penalties help under this benchmark's failure modes. A reader should treat this as a controlled engineering audit, not as a universal recipe.

Claim C4 is the adaptive-gate claim. It is explicitly pilot-label. The gate uses synthetic evaluator evidence to learn when raw high-N selection is risky. That is useful for diagnosing whether risk signals can be converted into selection decisions, but it is not a deployment guarantee. The audit script fails if the positive paired deltas and confidence intervals are absent.

Claim C5 is the learned-lite calibration claim. It uses a CPU NumPy calibrator on held-out synthetic graph rows. The key measurement is rank alignment with real utility and safe selected-utility behavior. This is intentionally weaker than a learned world-model claim. A full learned simulator would need separate training, rollout, generalization, and calibration evidence.

Claim C6 is the coverage claim. It says the suite crosses multiple graph families, hidden-failure modes, and stress levels. It does not say the suite covers all graph world-model failures. Coverage is local but broad enough to avoid a one-condition anecdote.

Unsupported claims C7–C10 are not caveats buried in prose. They are explicit non-claims. Real-system validation is absent. Broad benchmark superiority is absent. Universal monotonic improvement from increasing N is contradicted by the point of the paper. Deployment safety is outside the evidence surface.

H Graph Stress Suite Design

The suite intentionally separates observed graph structure from true constraint structure. The imagined graph is not merely noisy; it has a systematic blind spot. This is the reason the phrase constraint shadow is more precise than generic misspecification. The selected future can look good because it is well-aligned with a visible graph while being poor under constraints that the imagined graph cannot see.

The five graph families cover different relational priors. Ring-chord graphs create shortcut opportunities. Chains make long-range effects difficult for a local score. Grids expose spatial aliasing. Random geometric graphs add irregular degree patterns. Clustered graphs create modular interactions where missing cross-cluster constraints can matter. None of these are real environments. Their value is that the failure can be inspected family by family.

The hidden-failure modes are also mechanistic. Omitted edges remove a true constraint from the imagined score. Shifted rest lengths make a visible edge look well satisfied under the wrong equilibrium. Delayed damping punishes futures that look short-term plausible but drift under the true dynamics. Topology aliasing changes which relational pattern the imagined graph thinks it is controlling. Long-horizon drift creates a mismatch between short-horizon score improvement and later utility.

The stress levels are not used to claim monotonic severity. They are used to test whether the same audit protocol survives mild, medium, and severe hidden constraints. A strong paper should not depend on tuning one particular severity to make a figure pretty. The V4 matrix therefore averages at $N = 64$ across the stress suite and exposes graph-family and hidden-failure structure.

I Why the Finite Law Is Support Machinery

The finite law is exact and useful, but V4 deliberately demotes it from paper identity to measurement machinery. It has the same abstract shape in many selection problems: sort finite candidates by score, account for ties, and compute expected selected utility under repeated sampling. If the paper were only this law, it would be vulnerable to duplicate-submission concerns.

The graph contribution starts when the score groups have a physical meaning. High-score groups correspond to futures that the imagined graph likes. Low utility in those groups means the imagined graph's visible constraints do not capture the true constraint system. The theorem explains how increasing N moves probability mass into those groups; the graph suite explains why those groups can be physically misleading.

The law also keeps the empirical analysis honest. It prevents the manuscript from saying vaguely that search is bad. Search is not bad in the theorem. Search concentrates on high-score groups. The utility consequence depends on the utility of those groups. This is exactly the distinction the paper needs: larger N is useful when high-score futures are high-utility and risky when the graph score's extreme tail is a constraint shadow.

J Selector and Repair Threat Model

Raw selection is the attack surface. It chooses the candidate with highest internal graph score. In an ideal graph world model this is reasonable: the score is a compact proxy for future utility. In the stress suite, raw selection is intentionally vulnerable because hidden constraints are absent or distorted in the imagined graph.

Random selection is not a serious controller, but it is a necessary sanity baseline. It asks whether raw selection is meaningfully better than not using the score. Oracle selection is also not deployable. It provides an upper bound under the true evaluator and allows the paper to measure oracle regret. The paper should never present oracle as an available policy.

Observed-energy penalties ask whether visible graph strain is a useful warning. Action penalties ask whether extreme controls are part of the failure. A calibration-gap penalty asks whether imagined and true target errors diverge. Graph-energy calibration combines score and visible-graph diagnostics. Constraint probes use a hidden-risk proxy inside the synthetic suite. Combined repair tests whether these signals work better together than separately.

The adaptive gate is a stronger but more fragile control. It can decide not to use raw high-N selection when risk is high. Its advantage is operational: it changes which selector is deployed. Its weakness is informational: the gate uses pilot-label style evidence in the synthetic suite. V4 presents it as a diagnostic control, not a field-ready safety mechanism.

K Negative Controls

The first negative control is unsupported claim tracking. The repository contains claims that are deliberately marked unsupported. This is not a cosmetic exercise. It means the paper can be audited for overclaiming in the same way it can be audited for missing figures or tests.

The second negative control is the oracle. Oracle performance defines how much utility is left on the table, but it is never treated as a deployable method. If a paragraph implies oracle availability, that paragraph should be revised.

The third negative control is the learned-lite caveat. The learned model has held-out synthetic evidence, but it is small and feature-based. A reviewer should not accept any sentence that upgrades it into evidence for large neural world models. V4 uses it to show diagnosability, not broad learning success.

The fourth negative control is external validity. The paper repeatedly states that real robots and external benchmarks are absent. This reduces rhetorical strength but increases submission honesty. A scoped paper can be strong; an overstated synthetic paper is easy to reject.

L Statistical Rigor

The core paired comparisons use the high-N setting because the paper is about selected-tail behavior. V4 reports both all paired cases and hard cases. All cases ask whether repair or gating helps across the full $N = 64$ audit surface. Hard cases ask whether the same controls help when raw selection has a nontrivial oracle gap.

The confidence intervals are bootstrap intervals from the generated statistical table. V4 does not need a new statistical method; it needs clear reporting of which population the interval describes. The all-case adaptive gain is 0.00701 with lower endpoint 0.00529. The hard-case gain is 0.01447 with lower endpoint 0.01119. The V4 audit fails if these lower endpoints are not positive.

The exact-law validation is separate from the repair statistics. It measures whether the finite-pool accounting matches empirical selected utility. The mean absolute error 0.001341 is small enough to use the law as a diagnostic. It does not prove that the graph score is good, calibrated, or safe.

M Learned-Lite Calibration Audit

The learned-lite component is deliberately modest. It trains ridge regressors from synthetic candidate features to real utility and hidden energy. The feature set includes graph metadata and stress labels, so the result should not be read as hidden-constraint discovery from raw sensory data. It is an artifact-level question: are the failure signals in this suite linearly recoverable enough to improve rank alignment?

The answer is yes inside this artifact. Raw score rank correlation is 0.756, while learned utility rank correlation is 0.932. The learned-safe selector closes 0.588 of the hard-case oracle gap. Because the safe selector can still have negative deltas in a small fraction of cases, V4 avoids safety language and reports it as calibration evidence.

A reviewer could reasonably ask for a neural graph model trained from rollouts. That would be a stronger and different paper. This repository instead provides CPU-local calibration evidence that can be regenerated without large memory requirements. The result is useful because it demonstrates that selected-tail risk is not purely invisible, but it is not a substitute for a learned dynamics benchmark.

N Artifact Reading Guide

The result tables are the first source of truth. The main metrics table stores selector-level aggregates across N. The stress table exposes graph-family and hidden-failure structure. The repair and ablation tables explain which penalties are active. The adaptive-gate table records the gate decision surface. The learned-model table gives held-out calibration metrics. The statistical table gives paired deltas and confidence intervals.

The figures are summaries, not evidence replacements. Figure 1 is the narrative failure figure. Figure 2 checks stress-grid coverage. Figure 3 shows gating and oracle-gap closure. The V4 figures make the audit surface explicit: matrix, repair deltas, learned calibration, claim boundaries, artifact inventory, and family robustness.

The documents are guardrails. The claims document tells a reviewer what the paper is and is not allowed to claim. The final audit records which commands passed for v2. V4 adds a stricter build/audit script that checks source-map state, PDF hashes, page count, claim thresholds, and LaTeX blockers.

The source package is small enough to inspect. The graph-physics module defines the synthetic worlds. The selection module defines raw, oracle, random, repair, and gate behavior. The theory module implements the finite law. The learned module implements the CPU calibrator. The audit module enforces claim discipline.

O Non-Duplicate Identity Firewall

This paper must remain distinguishable from other selected-tail papers even if read side by side. Its object is graph constraint mismatch. It uses graph families, hidden physical constraints, graph-energy features, edge-violation proxies, topology aliases, and learned graph-feature calibration. Those details are not decorations; they are what the evidence measures.

A language-model selected-answer paper would have verifier scores and answer utility. A robot-policy EBM paper would have action energy and closed-loop policy evidence. A diffusion or trajectory paper would have denoising or sequence-generation structure. This paper has imagined versus true graph physics. Reviewers should not need the title to know that after reading the method and experiments.

The finite law is allowed to recur as a technical instrument because it is a shared measurement lens. The paper identity is not allowed to recur. If future edits make the first theorem the only memorable contribution, the manuscript should be rejected by its own audit. The V4 figures and appendix are designed to prevent that by making graph constraints impossible to miss.

P AREA-CHAIR DECISION SIMULATION

A skeptical area chair might say: the evidence is synthetic, but the paper is honest and well audited. That is a fair score. The likely accept argument is that graph world models can fail through constraint shadows, and the repository provides a clear finite-pool accounting law, a broad local stress suite, paired repair evidence, and explicit non-claims. The paper is useful as a diagnostic artifact.

The likely reject argument is that no external benchmark or real-system data is present. V4 cannot erase that weakness. It can only make sure the weakness is correctly scoped and that the local evidence is strong enough to be worth reviewing. A reviewer seeking benchmark novelty may still reject. A reviewer valuing controlled mechanism and artifact discipline may accept.

The desk-rejection risk is duplication. V4 reduces it by moving the manuscript away from generic selection language and toward graph-specific stress design. The title, abstract, figures, V4 section, and appendix all use the graph object. The theorem is still present, but it is framed as the accountant of the selected tail, not the full contribution.

Q RESOURCE AND REPRODUCIBILITY POLICY

The expensive experiment path is intentionally separated from the manuscript build. The bulletproof generation run records a long CPU-local runtime. V4 does not force every fresh session to rerun that path before compiling the paper. Instead, the cache script reads committed artifacts, generates compact V4 summaries and figures, and verifies that the manuscript uses fresh numbers.

This separation is important for RAM-light work. The V4 cache streams ordinary CSV/JSON files and uses small matplotlib figures. It does not allocate large simulator batches, train large models, or hold all candidate rollouts in memory. Full regeneration remains available through the existing scripts when a reviewer wants to rebuild the evidence from source.

The final audit is stricter than a normal build. It requires the V4 cache, the claim audit, the final repo PDF, the Desktop PDF, matching PDF hashes, at least 25 pages, the exact Desktop source-map row, absence of the old Desktop v2 PDF, and a clean LaTeX blocker scan. This is designed for future agents: they should not need this conversation to discover where the paper lives or which PDF is eligible.

R SUBMISSION-READINESS CHECKLIST

Before treating V4 as final, the following checks must pass:

1. Generate the V4 cache and figures.
2. Build the V4 PDF into the repository final directory and the Desktop.
3. Confirm the repo and Desktop PDF hashes match.
4. Confirm the final PDF has at least 25 pages.
5. Confirm the source map points to the V4 PDF, this folder, and the GitHub repository.
6. Run the original claim audit.
7. Run the V4 claim audit.
8. Run the Python test suite.
9. Check the LaTeX log for fatal errors, unresolved references, unresolved citations, and overfull boxes.

10. Render representative PDF pages and visually inspect text and figure placement.
11. Commit the final artifacts and push to GitHub.
12. Verify that the remote branch hash equals the local commit hash.

S REVIEWER ATTACK LEDGER

Attack: the paper is only a synthetic toy. Response: yes, it is a controlled synthetic mechanism paper. The paper should be judged on whether the mechanism, artifacts, and claim boundaries are valuable, not on evidence it does not claim to have.

Attack: the finite law is generic. Response: yes, the law is generic support. The contribution is the graph constraint-shadow audit built around that support.

Attack: the repair is tuned. Response: the repair coefficients are fixed synthetic controls, and the paper reports them as such. The stronger failure claim is independent of the repair being universal.

Attack: the adaptive gate uses labels unavailable online. Response: correct. The paper calls it pilot-label evidence and refuses deployment-safety language.

Attack: learned-lite calibration is too small. Response: correct for large-model claims. The supported claim is held-out rank alignment in the synthetic suite.

Attack: the graph families are hand-coded. Response: correct. The suite is a controlled benchmark object. External benchmark claims are marked unsupported.

Attack: high N is not always bad. Response: the paper agrees. The theorem and experiments say high N is good when high-score groups have high utility and risky when high-score groups are constraint shadows.

T FRESH-AGENT INVARIANTS

A fresh agent should be able to recover the final state without hidden context. The Desktop source map identifies the final PDF name, local folder, and GitHub repository. The repository contains the V4 final PDF, the V4 build script, the V4 audit script, the V4 cache, generated V4 figures, and the original evidence artifacts. The final PDF should not depend on transient local build files.

The final manuscript must remain anonymous and scope-honest. It should not leak local paths in the PDF text. It should not claim real-robot evidence, external benchmark superiority, or deployment safety. It should not make the finite law the sole contribution. It should present graph-specific stress design as the core.

The final repository must be pushed. A local-only PDF is not enough for this workflow because future sessions need the GitHub source of truth. Remote hash verification is therefore part of the completion condition.

U WHAT WOULD MAKE THE PAPER STRONGER LATER

The natural next step is an external physics benchmark where hidden constraints are not hand-authored by the paper. That would test whether constraint shadows appear in less controlled environments. The paper does not need that to make a scoped mechanism claim, but it would be the next evidence tier.

A second extension is a learned graph dynamics model trained from rollouts. The current learned-lite calibrator is a diagnostic model over synthetic features. A learned simulator would require separate train/test splits, rollout metrics, calibration tests, and failure analysis. It should not be slipped into the current claim without new artifacts.

A third extension is online gating without true-utility labels. The current gate is pilot-label. A deployable gate would need risk signals available at test time, thresholds fixed before evaluation, and evidence that it does not silently block useful high-N improvements.

These extensions are deliberately left outside V4. Adding them without enough evidence would weaken the paper by making it overclaim. V4 is strongest when it is precise: controlled graph constraint shadows, audited selected-tail repairs, and explicit boundaries.

V V4 EXTENDED AUDIT NOTES

V.1 PER-FAMILY THREAT MODELS

The ring-chord family is a shortcut test. A ring graph with chord edges gives the imagined model plausible short paths through the state space. If hidden constraints change which chord-like interactions matter, then the high-score future can satisfy the imagined shortcut while violating the true relation. A raw selector is vulnerable because it sees the shortcut bonus and visible-edge energy, not the hidden edge cost. The V4 matrix asks whether this vulnerability is an isolated drawing of one graph or a repeated family-level behavior.

The chain family is a propagation test. Chains are simple, but long-range effects must pass through many local interactions. A missed damping or rest length shift can be invisible at the local score horizon and still matter after selection. High N amplifies this because it finds futures that exploit the local scoring horizon most aggressively. The correct claim is not that chains are difficult in general; it is that a chain can expose score/utility mismatch when the imagined rollout and true rollout encode different propagation rules.

The grid family is an aliasing test. Regular local neighborhoods can make different physical configurations look similar to a graph score. Topology aliasing is especially relevant here because a candidate can appear consistent with one local pattern while violating another true pattern. The grid family therefore guards against the paper being only a shortcut story. It asks whether constraint shadows survive a more spatially regular relational structure.

The random-geometric family is an irregular-degree test. Random geometric graphs create uneven local neighborhoods. A score that treats visible local energy as sufficient can underweight missing constraints around high-degree or border nodes. This family is important because many learned graph models face irregular contact or proximity graphs rather than clean grids or chains.

The clustered family is a modularity test. Clustered graphs create strong within-cluster structure and weaker cross-cluster dependencies. Missing or misweighted cross-cluster constraints can be particularly deceptive: the visible score can look good inside each module while the true utility depends on between-module consistency. This family makes the constraint-shadow framing less dependent on one graph motif.

Taken together, these family threat models define the local benchmark identity. The paper is not claiming that these five families exhaust graph world models. It is claiming that they form a controlled, inspectable stress surface for the specific mechanism under study.

V.2 PER-FAILURE WALKTHROUGHS

Omitted edges are the cleanest hidden-constraint failure. The imagined graph does not know a true relation exists. A selected future can exploit the missing edge because the score never charges it. This is the most direct instance of a constraint shadow: the selected future is good inside the visible graph and bad under the complete graph.

Shifted rest lengths are subtler. The imagined graph sees an edge, but its preferred equilibrium is wrong. A future can keep visible energy low under the imagined rest length while accumulating true energy under the shifted one. This matters because real learned models often know that an interaction exists but misestimate the relation.

Delayed damping tests temporal mismatch. A future may look good during the selection horizon and then lose utility when damping appears later or with a different strength. This failure is important for world models because a short imagined rollout can be locally plausible while setting up long-horizon drift.

Topology aliasing tests relational identity. The imagined graph can confuse which topology is present or which local relation should be applied. The score therefore rewards a candidate that is appropriate for the wrong relational template. This failure is distinct from simply omitting an edge; it is about using the wrong graph semantics.

Long-horizon drift tests accumulated mismatch. The imagined and true graphs may agree at first but diverge as small errors accumulate. High-score selected futures can be especially vulnerable because they are extreme along the imagined score direction. The selected future may therefore be the one whose later drift is worst under the true evaluator.

These failure modes should be read as a taxonomy for a synthetic audit, not as a catalogue of all physical failures. Their purpose is to keep the paper from relying on one hidden bug. A stronger external paper would replace or augment them with failures induced by real learned dynamics.

V.3 Repair coefficient audit

Repair coefficients are fixed in the source package before the V4 manuscript is built. V4 does not retune them to create a new story. The coefficient audit is therefore about interpretation. Each penalty has a reason to exist, and each penalty has a scope boundary.

Observed-edge energy penalizes visible strain. It is useful when visible graph stress is correlated with hidden risk. It is weak when the hidden constraint is not reflected in visible edges. Action norm penalizes extreme controls. It is useful when the shortcut is created by aggressive action, but it can be unnecessarily conservative when high action is genuinely useful.

Calibration-gap penalties use disagreement between imagined and true target errors in the synthetic artifacts. This is not an online signal unless a separate estimator exists. In the current suite it is a diagnostic: if the gap helps repair selected utility, then the failure is at least partly visible to a calibration channel. The paper must not imply that this gap is freely available in deployment.

Constraint probes use a benchmark-specific hidden-risk proxy. Their purpose is to ask whether a small amount of extra graph-risk information can change the selected tail. This is useful scientifically because it distinguishes pure score failure from unobservable failure. It is not a claim that the probe is available in a real robot.

Combined repair merges several signals. It is the strongest repair baseline in the local audit because selected-tail failures can have multiple causes. Its success should be read as evidence that the failure is multi-signal and repairable inside this suite. Its failure in any condition would not refute the constraint-shadow diagnosis; it would identify a stronger repair target.

V.4 Alternative hypotheses

Alternative hypothesis one is random seed luck. The evidence counters this by using seed-level selector rows and paired comparisons. The V4 audit does not accept a single pretty trajectory. It requires aggregate rows, hard-case rows, and positive paired intervals for the adaptive claims.

Alternative hypothesis two is a one-family artifact. The V4 family and matrix figures counter this by crossing graph families and hidden-failure modes. A reviewer should still inspect the cells: uneven strength across families is expected and informative. The claim is cross-family coverage, not identical effect size everywhere.

Alternative hypothesis three is pure action magnitude. If action norm alone explained the failure, then graph-energy, calibration-gap, and constraint-probe features would be unnecessary. The ablation suite lets a reviewer compare single-component and combined repair. The paper does not need every component to dominate; it needs the decomposition to be visible.

Alternative hypothesis four is learned calibration leakage. The learned-lite model uses synthetic features and condition labels, so the concern is valid for external claims. V4 handles this by narrowing the claim to held-out synthetic rank alignment and by refusing to call the learned-lite result a general learned world model.

Alternative hypothesis five is that high N is simply too small. The suite uses N up to 64 because it is CPU-local and enough to expose selected-tail effects in the finite pools. A larger N sweep could be useful, but the paper's claim is not about asymptotic behavior. It is about the observable selection pressure in the generated candidate pools.

V.5 HARD-CASE INTERPRETATION

The hard-case subset is defined by raw oracle gap. This is not a hidden second benchmark; it is an analysis subset that asks where repair matters most. If raw selection already matches oracle utility, a repair cannot show much gain. Hard cases therefore prevent the average from being diluted by easy conditions.

The hard-case adaptive gain is 0.01447 with interval [0.01119, 0.01812]. This is the strongest repair number in the manuscript because it focuses on failures that actually need a repair. The all-case number remains necessary because a repair that helps hard cases by damaging easy cases would not be acceptable.

The paper should never use hard-case results alone to claim global improvement. The correct wording is that adaptive gating improves hard-case selected utility inside the synthetic audit, while all-case paired results check for broader damage. This wording is less dramatic, but it is what the evidence supports.

V.6 FAILURE-CASE RECONSTRUCTION PROTOCOL

To reconstruct a failure case, start with a row where raw $N = 64$ oracle regret is positive. Identify its graph family, hidden failure, stress level, and scenario. Compare raw selected score, selected real utility, hidden energy, edge violation, action norm, and calibration gap against oracle and repair selectors. The goal is not to find a single dramatic picture; it is to explain which hidden constraint the raw score failed to price.

The next step is to ask whether the repair feature that helps is physically meaningful. If observed-edge energy helps, the failure leaked into visible strain. If action penalty helps, aggressive controls were part of the shortcut. If calibration gap helps, imagined and true errors diverged. If constraint probe helps, hidden-risk information was necessary. If only oracle helps, the failure may be difficult to diagnose from available signals.

This protocol is important because it stops the paper from treating aggregate gain as magic. A reviewer should be able to move from a table row to a mechanistic explanation. V4 does not include every row-level walkthrough in the main text, but the appendix states the method and the repository exposes the tables needed for inspection.

V.7 FIGURE-LEVEL AUDIT NOTES

Figure 1 should be read first. It shows the selected-tail failure story in the simplest form. If that figure does not convince a reader that a score/utility gap exists, the rest of the paper will not rescue the claim.

Figure 2 should be read second. It asks whether the failure is spread across the stress grid. A sparse heatmap would make the paper an anecdote. A structured heatmap supports the claim that graph family and hidden failure interact with selected-tail risk.

Figure 3 should be read as a repair audit, not a new safety claim. It shows whether adaptive gating and repair close oracle gaps under the same synthetic evidence. It should be paired with the caveat that the gate uses pilot-label information.

Figure 7 is the V4 identity figure. It is where the paper visibly differs from a generic selected-tail paper. A reviewer comparing manuscripts side by side should see graph families and hidden failures here, not an interchangeable selection plot.

Figure 8 is the V4 statistical figure. It separates all cases from hard cases and compares combined repair, adaptive gate, and oracle. This prevents the paper from hiding which subset drives the strongest result.

Figure 9 is the calibration figure. Its job is narrow: demonstrate held-out rank alignment and selected-utility movement for the learned-safe rule. It does not promote a learned simulator claim.

Figure 10 and Figure 11 are audit figures. They exist to make claim boundaries, artifacts, and family robustness visible. They are not decorative and should remain in the final version unless replaced by stronger graph-specific audits.

V.8 TABLE-LEVEL AUDIT NOTES

Table 1 is the compact headline table. It must remain consistent with the generated CSV/JSON artifacts. If any headline number is edited manually, the paper becomes less trustworthy. V4 therefore adds macros for the new audit values and checks the generated summary.

Table 2 is a freshness table. Its purpose is to show that V4 is tied to the current artifact package. It reports the supported claims, unsupported boundaries, condition counts, row counts, exact-law error, and generated figures. It is not a substitute for the result tables; it is a reader-facing manifest.

A possible future table would list every graph-family by hidden-failure cell. V4 uses a figure instead because the matrix is easier to inspect visually. The CSV is still written by the cache script so a reviewer can audit exact values.

V.9 CLAIM WORDING RULES

Use “controlled synthetic graph physics” when describing the evidence. Do not use “robot validation” or “real-world validation.” Use “pilot-label gate” when describing the adaptive gate. Do not use “safe controller” or “deployment guarantee.”

Use “learned-lite calibrator” or “CPU NumPy calibrator” when describing the learned result. Do not use “large learned world model” or “neural simulator validation.” Use “held-out synthetic rank alignment” for the statistical claim.

Use “constraint shadow” for the central failure. It names the graph-specific mechanism: the visible graph casts a score surface that hides true constraints. Do not replace it with generic phrases like “more samples can be bad” unless the graph mechanism is also present.

Use “oracle regret” as an analysis quantity. Do not imply oracle selection is deployable. Use “raw selector” for the actual vulnerable selection mechanism. Use “combined repair” and “adaptive gate” for scoped controls.

V.10 AUDIT SCRIPT CONTRACT

The V4 audit script is intentionally strict. It regenerates V4 cached evidence and runs the existing claim audit. It checks supported and unsupported claim counts, condition counts, row counts, hard-case counts, exact-law error, adaptive and repair gains, learned calibration, family coverage, hidden-failure coverage, artifact inventory size, generated figure count, and summary CSV row counts.

The script also checks the final PDF as an artifact. It requires the repository final PDF and the Desktop PDF to exist and have identical SHA-256 hashes. It requires at least 25 pages. It requires the old Desktop v2 PDF to be absent. It requires the Desktop source map to point to the V4 PDF, this local folder, and the GitHub repository.

Finally, the script checks the LaTeX log. Fatal errors, emergency stops, LaTeX errors, undefined controls, unresolved references, unresolved citations, and overfull boxes are blockers. Underfull boxes are tolerated because the ICLR template, bibliography URLs, and small figures can create harmless underfull warnings.

V.11 SOURCE MAP CONTRACT

The Desktop source map is part of the workflow, not a note to self. Future sessions start from the visible Desktop PDFs and need to find the correct local folder and GitHub repository immediately. For this paper, the final source-map row must name the V4 PDF, the local graph world model folder, and the graph-world-model GitHub repository.

The source map is outside the repository, so the Git commit cannot protect it. That is why the V4 audit script reads it directly. A pushed repo without a correct Desktop row would still fail the workflow because a fresh agent could pick up the wrong PDF.

The old Desktop v2 file should be absent after finalization. Keeping both v2 and V4 visible makes it too easy for a future session to open the wrong paper. The final paper must be the only eligible graph world model PDF on the Desktop.

V.12 Comparison to a full external benchmark paper

A full external benchmark paper would need different evidence. It would define train/test environments, external simulators, learned graph dynamics baselines, and held-out tasks. It would report rollout prediction, planning utility, calibration, robustness, and ablations under fixed protocols. It would likely require more compute and more dependencies.

This paper is not that. It is a controlled mechanism and audit paper. It asks whether selected-tail optimization can exploit a mismatch between imagined and true graph constraints. The value is that the mechanism can be seen, measured, repaired locally, and bounded by claims. The weakness is that the mechanism is hand-authored.

The two papers would be complementary. The current paper can motivate what to measure in an external benchmark: raw high-N oracle regret, hidden-energy diagnostics, graph-family sensitivity, repair deltas, adaptive gate failures, and learned calibration. An external paper could then ask whether those metrics matter outside the synthetic suite.

V.13 Potential reviewer scores

A strong reviewer who values controlled mechanisms might give a positive score: the paper is narrow but honest, the graph-specific failure is clear, the artifacts are complete, and the claims are enforced. That reviewer may still ask for better external validation, but may accept the paper as a diagnostic artifact.

A reviewer who values benchmark scale may give a borderline or negative score: the paper lacks external physics benchmarks and real robots. V4 cannot fully answer that. The rebuttal should not pretend otherwise. It should point to the controlled mechanism and explain that external validation is future work.

A reviewer who worries about duplication should be directed to the graph identity: graph families, hidden constraints, repair features, learned-lite graph calibration, and family/failure matrices. The finite law is shared support, but the paper is not a wrapper around the law.

A reviewer who worries about overclaiming should find the paper unusually easy to audit. Unsupported claims are explicit. The V4 audit enforces them. The limitation section is not defensive; it is part of the paper's correctness.

V.14 Camera-ready risk register

Risk one is page pressure. If a future venue imposes a shorter main paper, the appendix can carry most V4 audit content. The graph-specific main section and figures should remain because they prevent duplicate-wrapper perception.

Risk two is dependency drift. The current build uses MiKTeX and standard Python packages already used by the repository. The cache is deterministic, but external TeX installations can still differ. The final PDF is committed so a fresh session has an exact artifact even if local TeX changes.

Risk three is stale evidence. If experiments are regenerated, the V4 cache and manuscript must be rebuilt before submission. The audit script catches many stale-number failures by reading generated summaries and checking the final PDF hash path, but authors should still inspect changed tables.

Risk four is accidental claim inflation during editing. This is the most likely human failure. The manuscript should be scanned for real-world, benchmark, state-of-the-art, universal, and safety language before submission. Any such language needs either new evidence or removal.

V.15 FINAL REVIEWER SCORECARD

Mechanism clarity: strong. The paper defines a specific graph-world-model failure in which imagined visible constraints differ from true constraints and high-score selection concentrates on the wrong tail.

Artifact rigor: strong inside the stated scope. The repository contains source, tests, generated tables, figures, claim maps, audits, and a final PDF. The V4 cache makes the evidence easier to inspect without rerunning the expensive path.

Empirical scope: moderate. The local synthetic suite is broad enough for a mechanism paper but not enough for a broad benchmark paper. This is the central remaining weakness.

Statistical reporting: solid. The paper reports paired deltas, confidence intervals, hard-case subsets, exact-law error, and learned held-out alignment. It should not be scored as a massive empirical benchmark.

Novel identity: strong after V4. The constraint-shadow framing, graph family audit, hidden-failure taxonomy, and graph-specific repair signals make the paper recognizable without relying on the generic finite law.

Claim discipline: strong. The paper says what it does not prove and keeps those non-claims machine-readable. This is an advantage, not a weakness, because it prevents reviewers from having to infer the boundaries.

V.16 AUTHOR-RESPONSE TEMPLATES

If a reviewer asks for real robots, the response should be: the paper agrees that real-system validation is absent and does not claim it. The contribution is a controlled graph constraint-shadow audit. We have clarified this in the abstract, limitations, claim audit, and V4 evidence section.

If a reviewer says the theorem is generic, the response should be: yes, the finite law is the accounting tool. The paper's graph-specific contribution is the stress suite and evidence around imagined versus true graph constraints, including family/failure matrices and repair/gate audits.

If a reviewer challenges the adaptive gate, the response should be: the gate is pilot-label synthetic evidence. We do not claim online safety. Its purpose is to show that risk signals in the suite can improve selected-tail behavior under paired high-N evaluation.

If a reviewer challenges learned-lite calibration, the response should be: the learned result is deliberately limited to held-out synthetic rank alignment. It does not claim large neural world-model validation. We report it because it tests whether constraint-shadow risk is diagnosable from candidate and graph features.

If a reviewer asks why no larger N appears, the response should be: the paper studies finite candidate-budget selection in CPU-local pools up to $N = 64$, where the selected-tail effect is already visible and audited. Larger budgets are a useful extension, but the current claim does not depend on asymptotic behavior.

V.17 EXPERIMENT PROTOCOL DETAILS FOR REVIEWERS

The experiment protocol has two layers. The first layer is generation. The generator builds synthetic graph conditions, samples candidate action sequences, rolls each candidate through the imagined graph and true graph, computes score features and utility features, and records selector outcomes across candidate budgets. This layer is expensive relative to normal manuscript compilation, so it is intentionally run only when regenerating the evidence.

The second layer is verification. Verification reads the generated artifacts and asks whether the claims still hold. The V4 cache belongs to this layer. It does not create new graph worlds; it reads the committed tables and builds summary artifacts that a reviewer can inspect quickly. This distinction matters because it lets the paper be both strong and practical: full regeneration is available, but ordinary submission checks stay small.

Every selector is evaluated on the same generated candidate pools. This paired design is what makes raw-versus-repair comparisons meaningful. If raw and adaptive gate were evaluated on different candidate pools, selected utility deltas would mix selector effects with sampling noise. The paired statistical table is therefore one of the most important artifacts in the repository.

The stress grid is generated before looking at V4 figures. V4 summarizes the grid; it does not choose the grid. This is important for avoiding hindsight cherry-picking. The V4 matrix can reveal uneven or weak cells, but it should not alter which families or hidden failures are included.

The learned-lite train/test protocol is also fixed before V4 summary. Its purpose is not to win a benchmark but to ask whether candidate features contain enough information to improve utility ranking on held-out synthetic rows. The safe selector threshold is part of that artifact and should not be retuned during manuscript editing.

V.18 Row-level audit questions

A reviewer can ask several row-level questions. For a raw high-N failure row, which hidden-failure mode is active? Which graph family is active? Does the selected candidate have high hidden energy, high edge violation, large action norm, or a positive calibration gap? Does combined repair choose a different candidate? Does adaptive gating block raw? Does oracle utility show that the candidate pool contained a better option?

The paper's aggregate claims should survive those row-level questions. If an aggregate gain comes from a few rows with no mechanistic explanation, the paper is weaker. If rows repeatedly show interpretable constraint-shadow signals, the paper is stronger. The repository is designed so this inspection is possible from CSV files rather than from hidden notebooks.

Row-level auditability also protects against narrative drift. A paragraph may sound plausible but be unsupported by rows. For example, saying that action penalty solves topology aliasing would need row-level or grouped evidence. If the table does not support that level of specificity, the manuscript should use broader wording.

V.19 Robustness questions left open

The first open robustness question is candidate-pool distribution. The current candidate sampler is deterministic and synthetic. A different sampler might produce different high-score tails. This does not invalidate the current mechanism, but it does mean the paper should not claim robustness to all proposal distributions.

The second open question is model learning. The imagined graph here is constructed rather than learned from data. A learned graph world model might show different errors: it may smooth hidden constraints, memorize some stress modes, or invent spurious relations. The current paper gives metrics that a learned model should be tested with; it does not prove learned-model behavior.

The third open question is adaptive threshold transfer. The gate's thresholds are defined inside the synthetic suite. It is unknown whether they transfer to new graph families or real systems. V4 therefore treats the gate as evidence that selected-tail risk can be acted on when labels or probes exist, not as a universal gate.

The fourth open question is long-horizon compounding beyond the chosen rollout length. Long-horizon drift is included as a hidden failure, but real systems can compound errors over far longer horizons. The paper's utility metrics are tied to the generated rollouts and should not be extrapolated to indefinite control.

The fifth open question is multi-object semantics. The graph families abstract relational structure but do not include perception, object identity uncertainty, contact discontinuities, or actuator delays from real robotics. Those would make the problem richer and would require new evidence.

V.20 Why V4 does not rerun bulletproof on every build

A naive submission workflow might rerun the full generator every time the paper is compiled. That would be expensive and unnecessary. The full generation path is for regenerating evidence. The build path is for checking that the current evidence is correctly represented in the paper. Conflating the two makes the workflow slower without making the final PDF more trustworthy.

V4 therefore uses a cache generator that is deterministic and small. It reads the existing tables, writes summaries, writes figures, and exports macros. If the underlying evidence changes, the cache changes. If the evidence is stable, the cache is stable. This is exactly the behavior needed for a submission package that must be pushed to GitHub and re-opened by future agents.

The full bulletproof path remains part of the artifact. A reviewer or author can run it when they want to regenerate the tables. But the final audit focuses on whether the committed evidence and PDF are internally consistent. That is the right finalization check because the final PDF is what will be submitted and the repository is what will be shared.

V.21 Boundary-preserving edits

Future edits should preserve three boundaries. The first boundary is evidence-source boundary: generated tables are evidence; prose is not evidence. If prose needs a number, it should come from a table, JSON file, or generated macro. Manual number edits create stale-paper risk.

The second boundary is method boundary: raw, oracle, repair, gate, and learned-safe selectors have different availability assumptions. Oracle is an analysis upper bound. Gate is pilot-label. Learned-safe is synthetic calibration. Raw and repair are score-based selectors inside the benchmark. Merging these assumptions in prose would confuse the contribution.

The third boundary is scope boundary: synthetic graph physics is the evidence domain. External simulators, real robots, and safety-critical deployment are not in the evidence domain. The paper can motivate those settings, but it cannot claim results there.

V.22 Adversarial reading order

An adversarial reviewer should read the abstract first and check whether it overclaims. Then they should inspect Table 2 to see the evidence surface. Then they should inspect Figure 7 to decide whether the graph constraint-shadow identity is visible. Then they should inspect the claim audit and limitations to see whether unsupported claims are explicit.

Only after that should the reviewer read the theorem. This reading order is intentional. If the theorem comes first mentally, the paper looks generic. If the graph evidence comes first, the theorem becomes a useful accounting tool.

Finally, the reviewer should run the audit commands. A paper package that cannot rebuild its PDF, regenerate its summaries, pass its tests, and verify its claim map is not submission-ready. V4 treats those operational checks as part of the scholarly artifact.

V.23 Open-review questions and traceability

The most useful open-review question is not whether the paper could be larger. It could. The useful question is whether each supported claim can be traced to a stable artifact. C1 traces to the theorem implementation, tests, and exact-law table. C2 traces to stress and main metrics plus selected-tail figures. C3 traces to repair, ablation, and paired statistical tables. C4 traces to adaptive-gate rows and paired intervals. C5 traces to learned-model metrics and predictions. C6 traces to the stress grid and run summary. If a claim cannot be traced this way, it should not be supported.

A second open-review question is whether the synthetic mechanism is still interesting after all caveats are applied. V4's answer is yes: graph world models are often trusted because relational structure appears physically meaningful, but a relational score can still hide missing constraints. The paper gives a compact way to audit that risk. The caveats reduce external scope, but they do not remove the local mechanism.

A third question is whether the result depends on the name "constraint shadow." It should not. The name is useful because it is memorable and graph-specific, but the evidence must stand without it. The actual object is the selected high-score tail under an imagined graph whose visible constraints do not match the true evaluator. If a reviewer dislikes the phrase, the data and claim boundaries remain.

A fourth question is whether the repair evidence should be in the main contribution. V4 keeps it because a failure-only paper is easier to dismiss as diagnostic but not actionable. The repair evidence shows that visible graph features, calibration gaps, and pilot labels can reduce selected-tail mismatch. The paper still keeps the repair claim smaller than the failure claim because the controls are engineered for the synthetic suite.

A fifth question is whether the artifact is too complex for the claim. V4's answer is that complexity is mostly in auditability. The core mechanism is simple, but the repository includes enough tables, figures, scripts, and claim maps to make the mechanism hard to misread. That is a feature for submission: reviewers can disagree with scope, but they should not have to guess what was run.

Traceability also protects future revisions. If an author adds an external benchmark, it should become a new claim tier with new artifacts, not a sentence inserted into the current scope. If an author adds a learned graph simulator, the learned-lite claim should either remain separate or be replaced by a new learned-model claim with its own tests. If an author changes the gate, the paired statistical table must be regenerated. The paper should evolve by adding evidence surfaces, not by stretching old ones.

V.24 Final pre-submission self-attack

Before submission, the author should try to reject the paper in four ways. The first rejection is "synthetic only." The correct response is to verify that every strong sentence says synthetic graph physics and that the title, abstract, limitations, and claim audit do not imply more. If any sentence overreaches, remove it.

The second rejection is "duplicate wrapper." The correct response is to read only the figures and section headings. If those do not clearly say graph constraint shadows, hidden graph failures, graph-family stress, repair signals, and learned graph-feature calibration, revise them. The paper must look different before a reviewer reaches the theorem.

The third rejection is "repair not trustworthy." The correct response is to separate repair from diagnosis. The failure mechanism should remain valid even if a reviewer distrusts the adaptive gate. The repair claim should be scoped to paired synthetic high-N evidence and should not carry the whole paper.

The fourth rejection is "artifact not reproducible." The correct response is operational: run the V4 build, V4 audit, claim audit, tests, determinism checks, visual render checks, and GitHub remote verification. If any fails, the paper is not final. Submission readiness is not a feeling; it is a set of artifacts that agree with each other.

V.25 Final acceptance conditions

The graph paper is final only if the following are simultaneously true. First, the PDF is at least 25 pages and contains graph-specific V4 content rather than padding. Second, the source map points to the V4 PDF and correct repository. Third, the repo and Desktop PDFs are identical. Fourth, the old Desktop v2 PDF is absent. Fifth, tests and claim audits pass. Sixth, generated V4 evidence is deterministic. Seventh, representative rendered pages look correct. Eighth, the GitHub branch contains the final commit.

If any condition fails, the paper is not finished. A pretty PDF is not enough. A pushed commit is not enough. A correct source map is not enough. This workflow treats the paper, repository, Desktop artifact, and GitHub remote as a single submission package.

W LLM Usage Statement

This draft manuscript was prepared with assistance from an LLM-based coding and writing agent under user direction. The agent was instructed to restrict claims to existing repository artifacts, cite only verified references, and preserve the synthetic scope of the evidence. Authors remain responsible for independently checking the manuscript, bibliography, generated PDF, and all submission metadata before any conference submission.

X Operational Rebuild Commands

The anonymous submission PDF can be rebuilt from the repository root with:

```
python scripts/build_v4_paper.py
```

The final package audit is:

```
python scripts/run_v4_claim_audit.py
```

The unit tests are:

```
python -m pytest -q
```

These commands intentionally reuse the committed CSV, JSON, and figure artifacts for low-RAM verification. The expensive synthetic generator remains available, but it is not required for rebuilding or auditing the submitted PDF.